

Do Coverage and Mutation Scores of LLM-Generated Test Suites Correlate with Their Effectiveness?

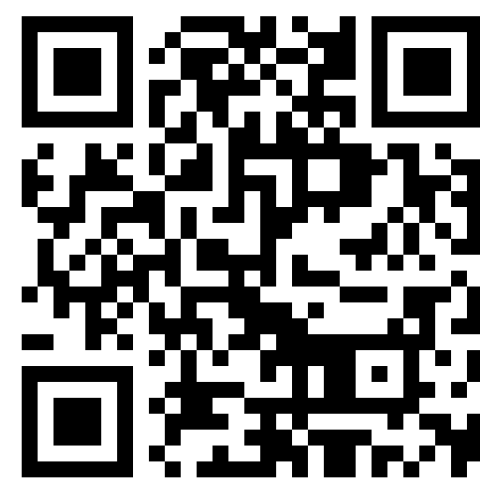
(Replicability Study)

ACM SIGSOFT DISTINGUISHED PAPER AWARD



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Paper (arXiv)



Replication package

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On the job market for postdoc positions
Happy to chat, open to questions.

8,268 suites · 101,123 tests · 318 focal methods from Defects4J, each with buggy and fixed versions · 13 configurations of 11 LLMs



PRIOR WORK · HUMAN-WRITTEN TEST SUITES

[Inozemtseva 2014] [Papadakis 2018]

- Correlations can largely vanish once suite size is controlled
 - Suite size: strong positive correlation with mutation score and bug detection
- [Inozemtseva 2014]
- Coverage types: all Pearson r above 0.95
 - "use simpler coverage measures"

LLM-GENERATED TEST SUITES

Bug-free input

1 Coverage and mutation can still be indicative of bug detection

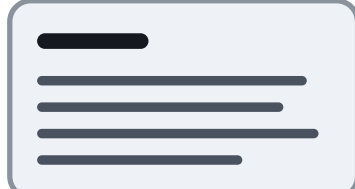
On bug-free input, coverage and mutation can be indicative of bug detection across models

- All models pooled: weak to moderate
- Within a model: weak

Three views

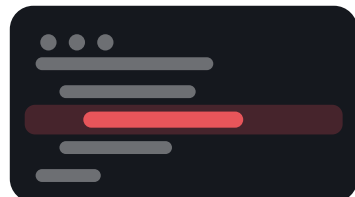
all models pooled · 1,000 draws of 100 suites per model
within a model · one model at a time
across models · one point per model

human-written tests

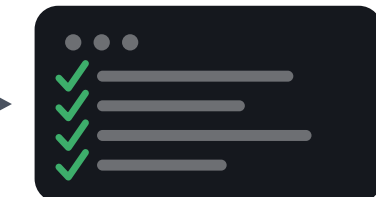


often grounded in developers' understanding of the intended behavior

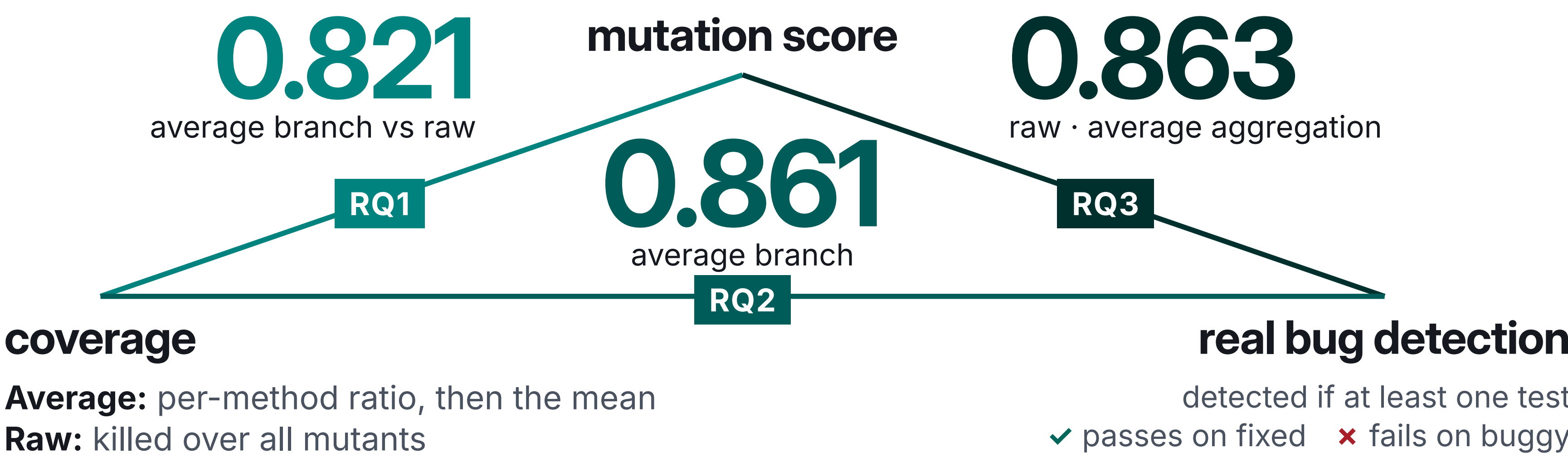
LLM-generated tests



code-under-test: the primary prompt input



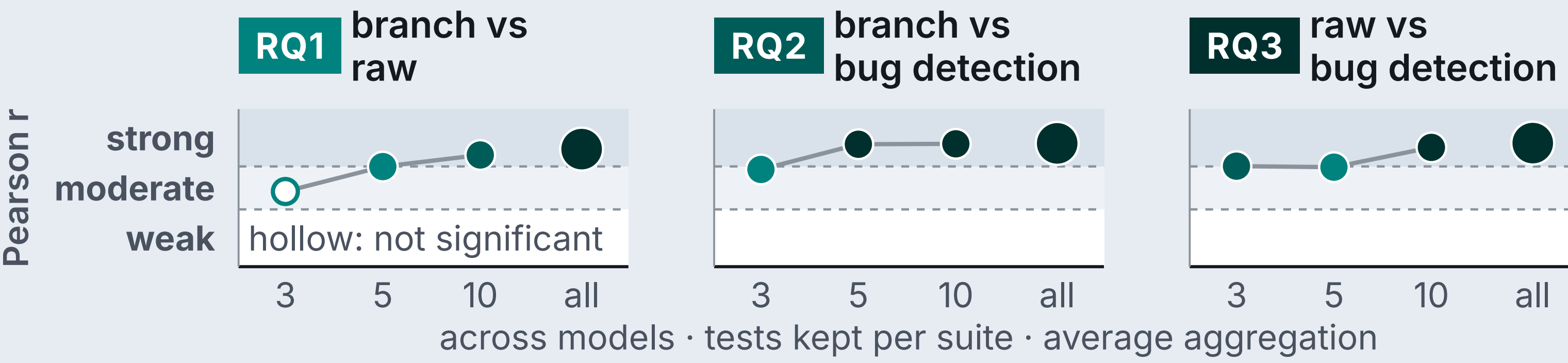
across models



2 Suite size: weakly correlated with mutation score, bug detection

Weak correlations observed in all views

Size fixed: qualitative strength largely retained



How test suites are constructed

PRIOR WORK sampled from a project-wide pool LLM-GENERATED one suite per focal method

Size likely reflects how many tests the focal method's behaviors need

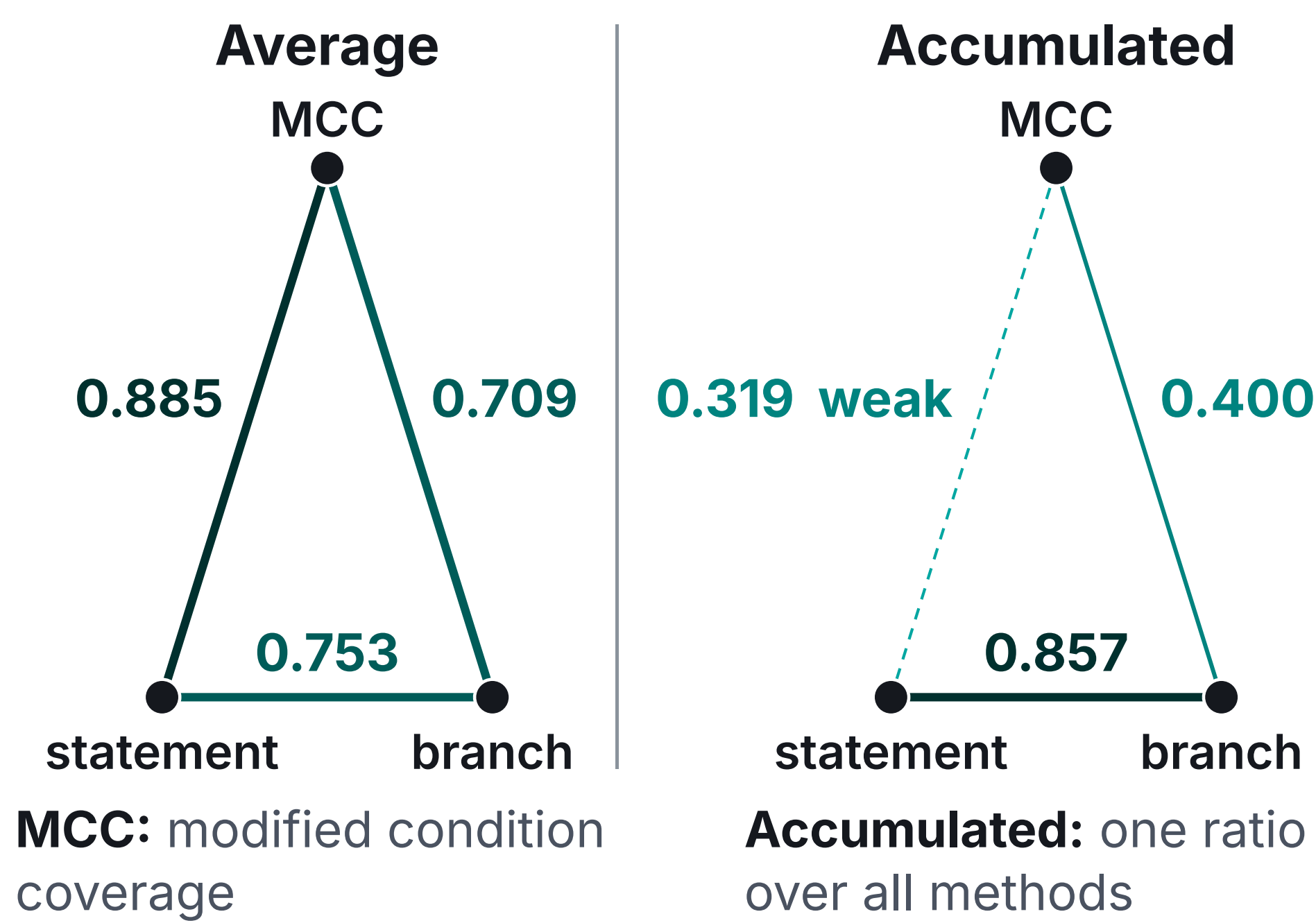
No strong confounding effect observed

3 Coverage types: related, not interchangeable

- Weaker than Inozemtseva et al. reported

Report diverse coverage types

all models pooled



Buggy input

4 The strong correlation turns weak

Measure bug detection directly

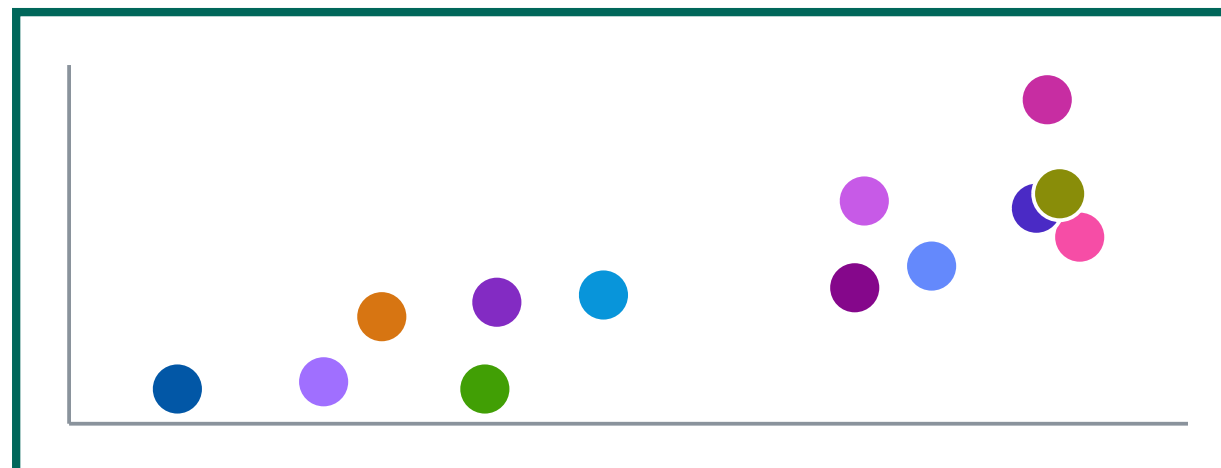
- Tests from buggy code can **encode faulty behavior**
- Mutation score: **not applicable**

Companion paper, ISSTA 2026:

Evaluating and Mitigating the Misguidance Effect of Buggy Code in LLM-Generated Unit Tests

BUG-FREE

r = 0.861



across models · average branch

BUGGY r = 0.35, not significant

